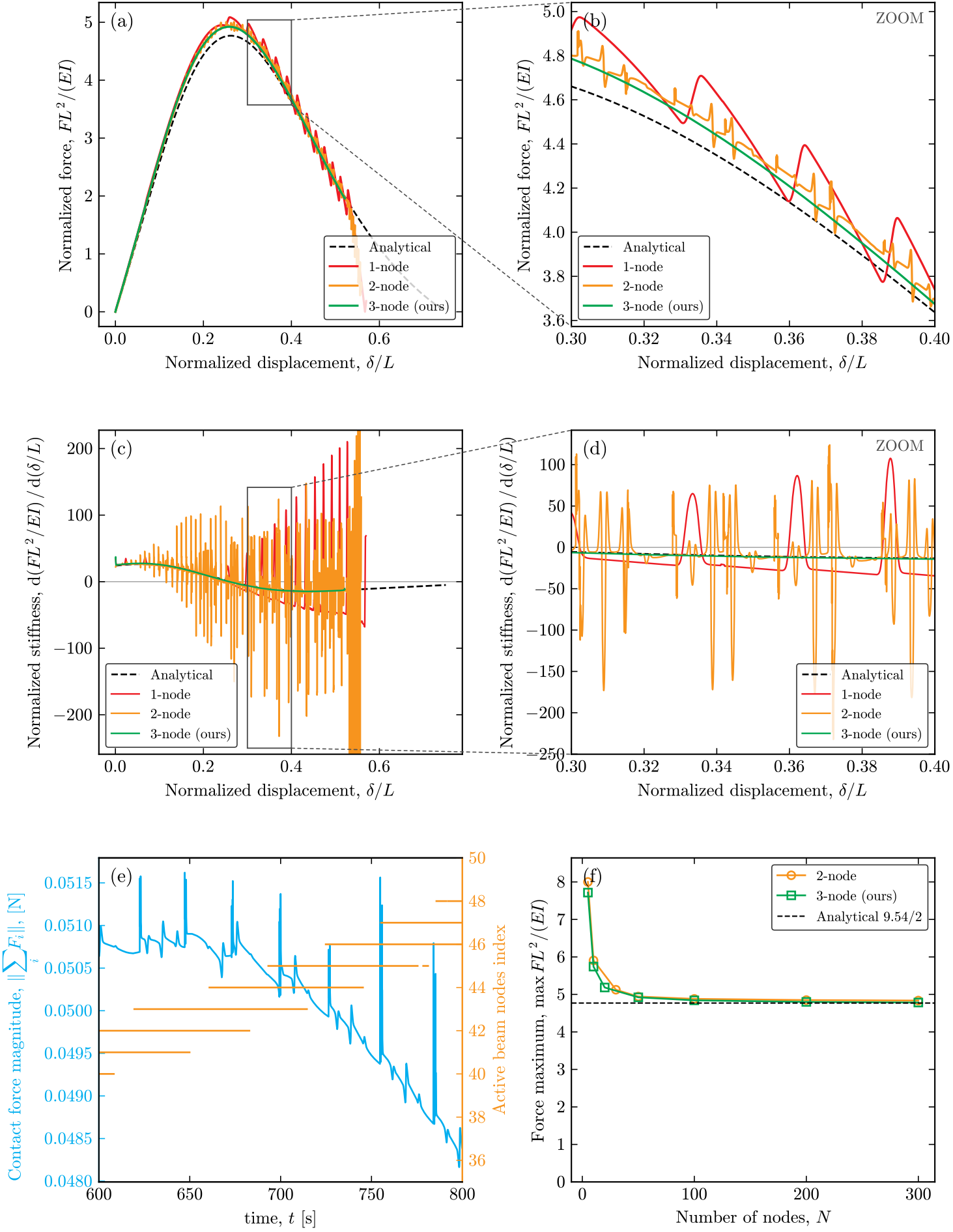

(a)
(b)
ZOOM
(c)
(d)
(e)
(f)
Analytical
1-node
2-node
3-node (ours)
Normalized force, $FL^2/(EI)$
Normalized displacement, δ/L
Normalized stiffness, $\mathrm{d}(FL^2/EI)/\mathrm{d}(\delta/L)$
Contact force magnitude, $\|\sum_i F_i\|$, [N]
Active beam nodes index
time, t [s]
Force maximum, max $FL^2/(EI)$
Number of nodes, N
Analytical 9.54/2